# SQL Review Research Activity

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## Part 1: SQL Fundamentals Review

1. What is SQL?

SQL or Structured Query Language, is the standard programming language used to interact with relational databases. It allows you to store, retrieve, update, and manage data in systems like MySQL, PostgreSQL, and SQL server.

2. Why is SQL important in database systems?

   SQL is important because it:

   - Provides a standard way to interact with databases
   - Allows efficient data retrieval (ex: searching/filtering large datasets)
   - Enables data manipulations (ex: insert, delete, update)
   - Controls access and security

3. Explain the difference between:
   - DDL (Data Definition Language) : used to define and modify database structure (commands: CERATE, ALTER, DROP)
   - DML (Data Manipulation Language) : used to manage or manipulate data inside the tables (command: INSERT, UPDATE, DELETE, SELECT)
   - DCL (Data Control Language) : used to control access/permissions. It controls who can access the data and what actions they are allowed to perform
   - TCL (Transaction Control Language) : it is a sequence of one or more DML statements that are treated as a single unit of work. It ensures the “All or nothing” principle, meaning either the whole sequence succeeds or none of it is applied (commands: COMMIT, ROLLBACK, SAVEPOINT)

4. Describe the purpose of the following SQL clauses/commands:
   - SELECT: used to retrieve data from the database
   - WHERE: used to filter datasets nbased on the conditions
   - ORDER BY: used to sort the results whether you want to from 1 to 10 or from 10 to 1
   - GROUP BY: used to group rows with similar values

- HAVING: used to filter the groups (used after GROUP BY) very similar to WHERE
- JOIN: used to combine rows from two or more tables based on a related column between them

1.
    a.
        -
        -
        -
        -
    b.
        -
        -
        -
        -
    c.
        - Retrieve a customer’s transaction history
        - Check account balances
        - Find transactions above a certain amount
2.
    a.
        -
        -
        - Orders and payment records
        - Shipping and delivery details
    b.
        - Manage product inventory
        - Process customer orders
        - Track sales performance
        - Analyze customer purchasing behavior
    c.
        - Display best-selling products

- View customer purchase history
- Update stock after a purchase

3.
   a.
      - Student records (name, ID, course)
      - Departments
      - Courses
      - Teacher information
      - Class schedules
      - Grades and attendance records

   b.
      - Store student academic records
      - Manage class enrollment
      - Track attendance and performance
      - Generate academic reports

   c.
      - Retrieve student grades
      - Calculate average scores by subject
      - List students enrolled in a course

# Part 4: Reflection

1. Which SQL concept do you find most challenging? Why?

   I find JOINs the most challanging because they required how efficient tables are related. When multiple tables are involved, it can be easy to make mistakes such as getting duplicate data or incorrect results if the join condition is not written properly.

2. Which SQL concept do you believe is most important for industry use? Why?

   I think SELECT with WHERE and JOIN is the most important. In real-world applications, most tasks involve retrieving specific data and combining information from multiple tables. These commands are used frequently in reporting, dashboards, and applications.

3. How does SQL support data-driven decision making?

SQL supports data-driven decision making by allowing organizations to quickly access, filter, and analyze large amounts of data. It helps identify trends, patterns, and insights that guide decisions.